

ENTERPRISE VIDEO/AUDIO CALLING PLATFORM - PRODUCT REQUIREMENTS DOCUMENT

Version: 1.0 | Date: 8/17/2026 | Author: Senior Architecture Team | Status: DRAFT

1. EXECUTIVE SUMMARY & CONTEXT

1.1 Overview & Problem Statement

Project: Enterprise-Grade Video & Audio Calling Platform

Scope: A distributed, microservices-based real-time communication platform enabling video and audio calls across desktop and web platforms, with integration capabilities for third-party applications.

Target Market: 400,000 concurrent and registered users across enterprise, SMB, and consumer segments requiring reliable, secure, low-latency communication.

- Current State: Market fragmentation with platforms requiring custom integration - no unified, modular solution exists
- Target Change: Provide a composable, SDK-first communication platform which can embed into other applications

1.2 Target Audience & User Personas

Primary User Persona #1: Enterprise IT Administrator

- All Cameroon state workers and citizens

End User Persona: Knowledge Worker

- Profile: Uses calling 3-5x daily for meetings, screen sharing, recordings
- Goals: Crystal clear audio/video, instant connection, intuitive UX
- Pain Points: Lag, call drops, poor mobile experience
- Success Criteria: <50ms latency, 99.9% uptime, mobile parity, one-click join

2. PRODUCT GOALS & SCOPE

2.1 Key Performance Indicators (KPIs)

Business KPIs:

- User Adoption: 400,000 registered users by Month 18 - Owner: Product
- Call Success Rate: 99.5% (calls established without disconnection) - Owner: Platform Eng
- Data Sovereignty over calls both audio and video in Cameroon

Technical KPIs:

- P99 Call Setup Latency: <1.5 seconds - Owner: Backend Lead
- Media Quality: Video bitrate 2-8Mbps (adaptive), audio 64-128kbps - Owner: Media Eng
- Uptime SLA: 99.9% (≤8.76 hours downtime/year per region) - Owner: DevOps
- Call Retention: 95% of calls > 1 min duration complete without involuntary disconnect - Owner: Infra

User Engagement KPIs:

- DAU/MAU Ratio: 40%+ (daily actives of monthly users) - Owner: Analytics
- Average Call Duration: >8 minutes - Owner: Product
- Feature Adoption: 60% use screen share, 40% use recording within first 30 days - Owner: Product

2.2 Feature Requirements

P0 (MVP - Must Ship) - Month 1:

- Core Calling:
 - 1-to-1 video & audio calls - no limit on call duration
 - Call initiation via room ID or direct peer invite
 - Desktop (macOS, Windows) and Web (Chrome, Firefox, Safari)
 - Adaptive bitrate (automatic quality adjustment based on bandwidth)
- User Experience:
 - Mute/unmute audio & video with keyboard shortcuts
 - Call state indicators (ringing, connecting, established, ended)
 - Call history with timestamps and duration
 - Basic contacts/recent calls list
- Security & Compliance:
 - End-to-end encryption (DTLS-SRTP for media)
 - TLS 1.3 for signaling
 - User authentication via JWT
 - GDPR-compliant data handling (explicit consent for analytics)

P1 (Phase 2- Month 6-9:

- Group Calling: 2-100 participant conferences (SFU architecture)
- Screen Sharing: Share desktop or application window
- Call Recording: Server-side recording with optional encryption
- Chat Integration: Text messages during calls
- Mobile (iOS/Android via React Native): Native apps for mobile users
- Advanced Analytics: Call quality metrics, participant behavior
- Webhooks: Call events (initiated, connected, recording started, ended)

P2 (Phase 3 - Nice-to-Have) - Month 12+:

- Virtual Backgrounds: AI-powered background blur/replacement
- Transcription: Real-time STT + full call transcripts
- Call Routing: Intelligent routing to available users
- PSTN Integration: Call to/from phone numbers
- Custom Branding: White-label portal for enterprise customers

2.3 Non-Goals (Out of Scope)

- SMS/Messaging Platform: This is communication calling only, not a messaging platform
- Video Conference Rooms: Not a Zoom replacement - focus on embedded SDK first
- Live Streaming: Different architecture, deferred to later phase

- VoIP PBX Features: Call queuing, IVR, hold music - enterprise PBX out of scope

3. UI/UX DESIGN SPECIFICATIONS

3.1 Design System & Component Library Guidelines

Design Philosophy: Minimal, distraction-free calling interface. Emphasis on clarity, accessibility, and reliability indicators.

Color Palette:

- Primary: #0066FF (Blue) - Used for: Active buttons, call indicators, highlights
- Secondary: #00AA44 (Green) - Used for: Active/connected states, record indicator
- Accent/Alert: #DD3333 (Red) - Used for: End call button, error states, warnings
- Neutral: #F5F5F5, #CCCCCC, #333333, #000000 - Text, backgrounds, borders

Typography:

- Display Font: Inter (sans-serif) - H1: 32px bold, H2: 24px bold, H3: 18px bold
- Body Font: Inter (sans-serif) - 16px regular (base), 14px small, 12px caption
- Monospace: Courier New/Monaco - For technical info, error codes
- Line Height: 1.5x for body text, 1.2x for headings

Spacing & Layout:

- Grid System: 8px base unit (all spacing in multiples)
- Standard Padding: 8px, 12px, 16px, 24px, 32px
- Call Interface: Full-screen video with fixed control bar (bottom, dark background)
- Border Radius: Small: 4px (buttons), Medium: 8px (cards), Large: 12px (modals)

Interactive States:

- Buttons: Default → Hover (opacity 0.8) → Active (pressed) → Disabled (0.5 opacity, no cursor)
- Icons: Muted indicators (red slash), recording (red dot pulse), connection status (spinner → checkmark)
- Audio Visualization: Real-time waveform during call, color-coded quality indicator

3.2 Responsive Screen Layouts

Desktop (1920x1080 and up):

- Layout: Full-screen video grid (1:1 aspect ratio), floating UI controls (bottom center)
- Call Controls: Mute/Unmute, Camera On/Off, End Call, Screen Share, Rec button
- Participant Panel: Thumbnails on right side (collapsible), participant names on hover

Laptop/Smaller Desktop (1440x900):

- Layout: Main video + smaller thumbnails in corner
- Controls: Same buttons, slightly reduced size
- Responsive: Auto-stack controls vertically if screen < 1200px

Tablet/iPad (768px-1024px):

- Layout: Portrait - full video + overlay controls, Landscape - side-by-side layout
- Touch Targets: Minimum 44x44px for all touch controls
- Controls: Larger buttons, haptic feedback on iOS

3.3 Component UI States

Call Screen States:

- Idle/Home: List of recent calls, quick-dial input, create room button
- Ringing (Incoming): Fullscreen caller info, accept/decline buttons, custom ringtone
- Connecting: Spinner + "Connecting..." message, don't allow user interaction
- Connected: Active video/audio, controls enabled, show participant info
- Media Error (No Camera/Mic): Fallback UI with warning icon, option to continue audio-only
- Network Poor: Quality indicator badge (red), auto-resolution adjustment notice
- Call Ended: Summary screen with duration, quality rating option, next actions

Participant Component States:

- Video On: Video stream visible, name overlay, connection quality indicator
- Video Off: Avatar/placeholder, mute badge, name visible
- Audio Muted: Mute icon overlay, gray tint
- Speaking: Active speaker highlight (subtle glow/border)
- Inactive: Dimmed, "no activity" indicator after 30s no audio

3.4 Accessibility Standards (WCAG 2.1 AA)

- Color Contrast: All text 4.5:1+ (large text 3:1+), icon states distinguishable without color alone
- Keyboard Navigation: All controls reachable via Tab, Space to activate, Arrow keys to select
- Screen Reader: Semantic HTML, ARIA labels for all buttons, live regions for call state changes
- Font Size: Minimum 16px for body text, user can zoom 200% without horizontal scroll
- Motion: Reduce animations on prefers-reduced-motion, no auto-playing videos
- Focus Indicators: Clear 3px outline on focused elements, high contrast (dark on light bg)

4. APP FLOW & USER JOURNEYS

4.1 Navigation & Site Map

Desktop App Structure:

- Home Screen
 - - Recent Calls List
 - - Contacts Directory
 - - Create New Call (button)
 - - Settings (profile, privacy, audio devices)
- Call Screen
 - - Full-screen video view
 - - Participant list (side panel)
 - - Call controls (bottom bar)
 - - Screen share preview (overlay)
- Settings/Profile
 - - Device management (camera, microphone, speaker)
 - - Privacy settings

- - Notifications
- - Recording preferences

Web App Structure:

- Login Page
- Direct Call Link: /call/[roomId] - Join directly without login

4.2 Primary User Journeys (Happy Paths)

Journey 1: Initiate 1-to-1 Call

- Step 1: User opens app, navigates to Contacts or Recent Calls
- Step 2: User selects contact to call
- Step 3: System shows "Calling..." with participant name, caller can cancel
- Step 4: Recipient receives notification (in-app + OS notification), can accept/decline
- Step 5: Both parties see video/audio streams, controls enabled
- Step 6: User can mute/unmute, toggle video, share screen during call
- Step 7: Either party ends call - return to home screen with call summary
- Time on Task: 5-10 seconds from contact selection to call established

Journey 2: Join Call via Link (No Login Required)

- Step 1: User receives link: <https://videocall.app/call/abc123>
- Step 2: Click link, browser opens (no download required for web)
- Step 3: Simple prompt asks for name, camera/mic permissions
- Step 4: User joins room and sees existing participants
- Step 5: Full calling experience available immediately
- Time on Task: 3-5 seconds from link click to seeing video

4.3 Edge Cases & Error Recovery Paths

- No Microphone/Camera Access: Show permission prompt, allow audio-only fallback, clear instructions
- Network Offline: Display "You're offline" banner, queue call state changes, auto-resume when online
- Poor Network (High Jitter/Packet Loss): Auto-reduce resolution, show "Network Poor" badge, suggest ending other apps
- Browser Not Supported: Show compatibility message with supported browsers (Chrome 90+, Safari 14.1+, Firefox 88+)
- Session Expired: Prompt re-login, preserve call state if possible
- Participant Dropped (Network): Show "Participant disconnected" for 30s, auto-remove if not reconnect
- Server Error During Call: Attempt graceful degradation (audio-only if video fails), show error code, offer callback
- Incoming Call While On Call: Show "Another call incoming" notification, allow transfer or decline

5. ASSUMPTIONS, RISKS, & DEPENDENCIES

Key Assumptions:

- Assumption 1: 400,000 users will have sufficient broadband capacity (avg 2.5Mbps download)
- Assumption 2: WebRTC will remain the de-facto standard for real-time communication (no major protocol shift)

- Assumption 3: STUN/TURN infrastructure available globally - we can deploy edge servers in major regions
- Assumption 4: Third-party integrations want SDK-first approach (not platform lock-in)
- Assumption 5: Regulatory compliance (GDPR, CCPA, HIPAA) is needed but standardizable

Known Risks:

- Risk 1: WebRTC NAT Traversal | Likelihood: MEDIUM | Impact: HIGH
 - Issue: 20-30% of users behind restrictive firewalls/symmetric NATs
 - Mitigation: Deploy global TURN infrastructure in 6 regions, fallback SFU relay server
- Risk 2: Media Quality at Scale | Likelihood: MEDIUM | Impact: HIGH
 - Issue: Bitrate, codec mismatches, packet loss > 5%
 - Mitigation: Implement adaptive bitrate (VP9, H.264), FEC, jitter buffers, extensive testing
- Risk 3: Vendor Platform Changes | Likelihood: MEDIUM | Impact: MEDIUM
 - Issue: Apple/Google/Mozilla change WebRTC APIs
 - Mitigation: Abstraction layer in SDK, monitor platform changelogs, early beta testing
- Risk 4: STUN/TURN Service Reliability | Likelihood: LOW | Impact: CRITICAL
 - Issue: Third-party STUN servers go down, TURN expensive at scale
 - Mitigation: Build internal STUN/TURN infrastructure, SLA-based vendor contracts

External Dependencies:

- WebRTC Libraries: Maintain own fork of libWebRTC or use vendor (Janus, Kurento)
- Codec Support: Ensure H.264, VP8, VP9, Opus codec support across platforms
- Certificate Authority: Let's Encrypt for TLS (auto-renewal), DTLS certs for SRTP
- Cloud Infrastructure: AWS/GCP for compute, Cloudflare/Akamai for edge TURN

CONFIDENTIAL - Enterprise Architecture Document